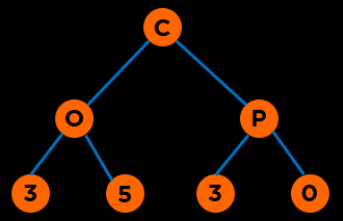
Exam 1 Review - A



Data Structures & Algorithm

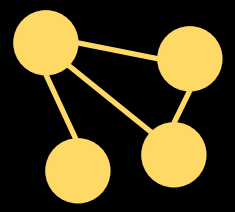
Categories of Data Structures

Linear Ordered

- Lists
- Stacks
- Queues

Non-linear Ordered

- Trees
- Heaps



Agenda

- Record this Lecture
- Announcements
- Exam 1 Logistics
- Exam Review A: Module 2 and 3

Exam 1 Logistics

- Date and timings :
  - For campus students (excluding students who need accommodations): Exam 1 will be on Feb 26 (Thursday) at 8:20 pm to 10:10 pm on campus in-person. The location of the exam is dependent on your last name.
  - For campus students who need accommodations as per DRC: Exam 1 will be on Feb 26 (Thursday). You can take the exam on Honorlock anytime between 6 pm to 11:59 pm at your home.
  - For UFOL/UDER students: Exam 1 will available between 8:20 pm on Feb 26 (Thursday) to midnight on Feb 28 (Saturday). You can take the exam on Honorlock anytime at your home.
- The exam duration is 1 hr 50 minutes hours.
- You are allowed the following for your exam:
  - one double sided handwritten sheet of notes (letter size/A4 page).
  - 4 double sided blank sheets of scratch paper (letter size/A4 page).

Exam 1 Logistics

Building and Room Number	Student Distribution (Last Name)
CLB Chemistry Laboratory C130	Aaron - Hirn
CSE Computer Sciences/En A101	Ho - Parks
NRN Norman Hall 1020	Patel - Zou

Exam 1 Topics and Expectations

https://docs.google.com/document/d/1XAUGHQE4JWmOsDkkbfTnLFoVi_QXySxC-J4XZnK4VTQ/edit?tab=t.0

Mini Review – Complexity

What is the computational complexity of adding an item to a stack in the worst case in terms of Big O notation?

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Answer: O(1)

Mini Review – Complexity

What is the computational complexity of the following code snippet?

```
int k;
for (int i=1; i < n; i++)
      for (int j=n; j>1; j=j/2)
            k = i*j;
```

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      for (int j=n; j>1; j=j/2)
            k = i*j;
```

Answer: $O(n \log_2 n)$

Mini Review – Complexity

What is the computational complexity of the following code snippet?

```
int i = 1;
while (i < n)
{
   i = i * m;
}
```

Mini Review – Complexity

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```
int i = 1;
while (i < n)
{
   i = i * m;
}
```

Answer: $O(\log_m n)$

Mini Review – Complexity

What is the computational complexity of the following code snippet?

```
result = 0
for (int i = 0; i < n; i++)
  result += i;
for (int j = 1; j < m; j *= 2)
  result *= j;
```

Mini Review – Complexity

What is the computational complexity of the following code snippet?

```
result = 0
for (int i = 0; i < n; i++)
  result += i;
for (int j = 1; j < m; j *= 2)
  result *= j;
```

Answer: $O(n+\log_2 m)$

Mini Review – Complexity

What is the computational complexity of the following code snippet?

```
for (int i=n; i>0; i/=2)
      for (int j=1; j<i; j++)
            sum = sum + 1;
```

Mini Review – Complexity

What is the computational complexity of the following code snippet?

```
for (int i=n; i>0; i/=2)
      for (int j=1; j<i; j++)
            sum = sum + 1;
```

Answer: O(n)

Mini Review – Complexity

Which code snippet will take less time to execute on a computer?

A

```
for (int i=1; i < n; i++)
      for (int j=1; j < n; j++)
            sum = sum + 1;
```

B

```
for (int i=1; i < 2n; i++)
      for (int j=1; j < 2n; j++)
            sum = sum + 1;
```

Mini Review – Complexity

Which code snippet will take less time to execute on a computer?

A

```
for (int i=1; i < n; i++)
      for (int j=1; j < n; j++)
            sum = sum + 1;
```

B

```
for (int i=1; i < 2n; i++)
      for (int j=1; j < 2n; j++)
            sum = sum + 1;
```

Answer: A

Mini Review – Complexity

Which code snippet will have a higher growth rate asymtotically in terms of Big O notation?

A

```
for (int i=1; i < n; i++)
    for (int j=1; j < n; j++)
        sum = sum + 1;
```

B

```
for (int i=1; i < 2n; i++)
    for (int j=1; j < 2n; j++)
        sum = sum + 1;
```

Mini Review – Complexity

Which code snippet will have a higher growth rate asymptotically in terms of Big O notation?

A

```
for (int i=1; i < n; i++)
    for (int j=1; j < n; j++)
        sum = sum + 1;
```

B

```
for (int i=1; i < 2n; i++)
    for (int j=1; j < 2n; j++)
        sum = sum + 1;
```

Answer: A and B grow at the same rate asymtotically

Mini Review – Linked Lists

Consider a class List that implements an ordered list backed by a singly linked list with a head pointer. The invariant “ordered” is maintained always. Given that representation, what is the worst-case time complexity of the following operations? Assume the list is sorted in ascending order.

A. Insert an item
B. Finding the minimum element
C. Delete the largest element from list
D. Finding the largest element
E. Finding a random element, n
F. Deleting the minimum element in the list

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A. Insert an item : O(n)
B. Finding the minimum element : O(1)
C. Delete the largest element from list : O(n)
D. Finding the largest element : O(n)
E. Finding a random element, n : O(n)
F. Deleting the minimum element in the list : O(1)

Mini Review – Stacks

Postfix Evaluation “2 3 1 * + 9 -“. We scan all elements one by one.

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1) Scan ‘2’, it’s a number, so push it to stack. Stack contains ‘2’
2) Scan ‘3’, again a number, push it to stack, stack now contains ‘2 3’ (from bottom to top)
3) Scan ‘1’, again a number, push it to stack, stack now contains ‘2 3 1’
4) Scan ‘*’, it’s an operator, pop two operands from stack, apply the * operator on operands, we get 3*1 which results in 3. We push the result ‘3’ to stack. Stack now becomes ‘2 3’.
5) Scan ‘+’, it’s an operator, pop two operands from stack, apply the + operator on operands, we get 3 + 2 which results in 5. We push the result ‘5’ to stack. Stack now becomes ‘5’.
6) Scan ‘9’, it’s a number, we push it to the stack. Stack now becomes ‘5 9’.
7) Scan ‘-‘, it’s an operator, pop two operands from stack, apply the – operator on operands, we get 5 – 9 which results in -4. We push the result ‘-4’ to stack. Stack now becomes ‘-4’.
8) There are no more elements to scan, we return the top element from stack (which is the only element left in stack).

Output Prediction / Coding Questions

https://onlinegdb.com/BJ4hyD7vP
https://onlinegdb.com/BJ6gewQDP
https://onlinegdb.com/OefiHIntV

Mini Review – Coding questions

- Check whether a string is a Palindrome using a Stack.
- Write C++ code for adding an element in the rear in a doubly linked list consisting of a head and tail.
- Write C++ code to find the median in a single linked using slow and fast pointer technique. – Edugator 4.5

Mini Review – Coding questions

- Design a Stack data structure in C++ that supports push, pop and min operations in O(1) time. You can use an STL Tack if needed.

Mini Review – Coding questions

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```
class MinStack {
    private:
        stack<int> s;        // main stack
        stack<int> minStack; // keeps track of current minimums

    public:
        // Push an element onto the stack
        void push(int x) {
            s.push(x);

            // If minStack is empty or x <= current min, push to minStack
            if (minStack.empty() || x <= minStack.top())
                minStack.push(x);
            else
                minStack.push(minStack.top());
        }
```

Mini Review – Coding questions

- Design a Stack data structure in C++ that supports push, pop and min operations in O(1) time. You can use an STL Tack if needed.

```
        // Pop the top element
        void pop() {
            if (s.empty())
                throw underflow_error("Stack is empty");
            s.pop();
            minStack.pop();
        }


        // Get the current minimum in O(1)
        int min() {
            if (minStack.empty()) {
                throw underflow_error("Stack is empty");
            }
            return minStack.top();
        }
};
```

Questions